Doping Stability Report -- Sc in LiCoO2 at 350 C

Generated: 2026-05-30 23:53   Host: LiCoO2   Dopant: Sc

Executive Summary

Sc preferentially occupies the Co layer. The formation energy difference between the two layers is +8.430 eV, strongly favouring the Co site. MD confirms this preference: on the Co site the dopant MSD plateau slope is 0.00435 A^2/ps (stable), compared to 0.00857 A^2/ps on the Li site. The Co-site result is isovalent (no charge mismatch), so the formation energy is fully reliable. The Li-site E_f is approximate due to a charge surplus of +2 that cannot be modelled in CHGNet.

Site	E_f (eV)	Charge	Compensation	MSD pass	Coord pass	Verdict
Co	-5.490	+0	No	PASS	PASS	STABLE
Li	+2.940	+2	Yes	FAIL	FAIL	UNSTABLE (migration + structu

Sc at the Co layer -- STABLE

Charge situation

Sc replaces Co with no change in oxidation state (isovalent substitution, charge mismatch = 0). No charge compensation is needed. The formation energy is fully reliable.

Formation Energy (thermodynamic stability)

Formation energy E_f	**-5.4904 eV**	PASS	*< 1.0 eV to pass*

E_f = -5.490 eV. This value is exceptionally large and negative, indicating that Sc incorporation is strongly thermodynamically driven at the Co site.

Molecular Dynamics Stability (350 C, 25 ps)

MSD slope (late-time)	**0.00435 A^2/ps**	PASS	*< 0.005 to pass (plateau = stable)*
Final MSD	**0.478 A^2**	----	
Max displacement	**0.796 A**	----	
Coordination (min/mean/max)	**6 / 6.0 / 7**	PASS	*relaxed = 6, tolerance +/-1*
Mean bond length (MD vs relaxed)	**2.058 A (relaxed: 2.0**	----	
Lattice volume change	**+0.00 %**	PASS	*< +/-5% to pass*

Late-time MSD slope = 0.00435 A^2/ps. The slope is small and below the migration threshold -- the dopant is site-stable. Final MSD = 0.478 A^2; maximum displacement from starting position = 0.80 A.

Coordination fluctuated between 6 and 7 (mean 6.0), within the +/-1 tolerance of the relaxed value (6). This is consistent with normal thermal vibration. Mean nearest-neighbour distance during MD = 2.058 A.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: STABLE

Sc is thermodynamically favorable and kinetically stable on the Co site at 350 C. All four stability criteria pass: formation energy is favorable, the dopant does not migrate during MD, its coordination environment is preserved, and the host lattice volume is unchanged.

Sc at the Li layer -- UNSTABLE (migration + structural distortion)

Charge situation

Sc replaces Li with a charge surplus of +2. CHGNet cannot model this compensation (would require Co oxidation or electron addition). The formation energy below is approximate.

Formation Energy (thermodynamic stability)

Formation energy E_f	**+2.9401 eV**	FAIL	*< 1.0 eV to pass*

E_f = +2.940 eV (approximate -- charge surplus not modelled in CHGNet). This value is large and positive, indicating that Sc incorporation is thermodynamically unfavorable under equilibrium conditions at the Li site.

Molecular Dynamics Stability (350 C, 25 ps)

MSD slope (late-time)	**0.00857 A^2/ps**	FAIL	*< 0.005 to pass (plateau = stable)*
Final MSD	**2.403 A^2**	----	
Max displacement	**1.958 A**	----	
Coordination (min/mean/max)	**4 / 6.1 / 8**	FAIL	*relaxed = 6, tolerance +/-1*
Mean bond length (MD vs relaxed)	**2.102 A (relaxed: 2.(**	----	
Lattice volume change	**+0.00 %**	PASS	*< +/-5% to pass*

Late-time MSD slope = 0.00857 A^2/ps. The slope exceeds the plateau threshold, suggesting mild drift or site-hopping. Final MSD = 2.403 A^2; maximum displacement from starting position = 1.96 A.

Coordination fluctuated between 4 and 8 (mean 6.1), deviating by more than +/-1 from the relaxed value (6). This may indicate structural distortion, or that the bond-length cutoff is too close to the actual bond length -- check the coordination_cutoff_A setting. Mean nearest-neighbour distance during MD = 2.102 A.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: UNSTABLE (migration + structural distortion)

Verdict: UNSTABLE (migration + structural distortion). Refer to the individual criteria above.

Site Preference Comparison

The Co site is preferred by 8.430 eV over the Li site. Formation energy: -5.490 eV (Co) vs +2.940 eV (Li).

Sc preferentially occupies the Co layer. The formation energy difference between the two layers is +8.430 eV, strongly favouring the Co site. MD confirms this preference: on the Co site the dopant MSD plateau slope is 0.00435 A^2/ps (stable), compared to 0.00857 A^2/ps on the Li site. The Co-site result is isovalent (no charge mismatch), so the formation energy is fully reliable. The Li-site E_f is approximate due to a charge surplus of +2 that cannot be modelled in CHGNet.

Caveats and Limitations

? **CHGNet is charge-neutral.**

The potential does not model oxidation states, polaron formation, or electron/hole localisation. Formation energies for aliovalent dopants (charge mismatch != 0) are approximate even when structural compensation defects are included.

? **No Freysoldt/Kumagai correction.**

Electrostatic finite-size corrections for charged defects require the host dielectric tensor and the DFT electrostatic potential, neither of which is available from a neural-network potential. For publication-quality E_f of charged defects, follow up with DFT using doped or pymatgen-analysis-defects.

? **Metal-rich reference state.**

Chemical potentials are computed from elemental ground-state structures. Synthesis under oxide-rich or other conditions shifts all E_f values by additive constants; the relative ordering of sites is unaffected.

? **MD timescales are short (25 ps default).**

This is sufficient to assess local site stability and rapid migration, but cannot resolve slow diffusion mechanisms. A non-plateauing MSD confirms instability; a plateau does not guarantee stability on longer timescales.

? **Energy rankings are more trustworthy than absolute values.**

CHGNet is a universal MLIP trained on r2SCAN DFT. Formation-energy errors for transition-metal oxides are typically 50-150 meV/atom. Use relative orderings (which site is preferred?) as the primary result.